

PROJECT DESIGN SCOPE AND TIMELINE OVERVIEW

This document outlines the scope, core objectives, and timeline for the avocado smartphone project. It defines the high-level goals and answers the fundamental question of what is actually being built. This serves as the definitive technical compass for the rest of the project, setting target performance standards, deadlines for the project and setting a finalized maximum budget.

1. Project scope

The scope of this project is large, as it includes designing most subsystems of a modern smartphone from scratch. However, certain subsystems will not be recreated from scratch but will instead be integrated as off-the-shelf components. This section defines which aspects of the project will be custom-built and which will be integrated as pre-existing elements.

1.1 - Off-the-shelf components

It is more efficient to start by defining what will not be custom-designed but rather purchased or integrated into the system layout without significant modification. This section details the technical justifications for selecting off-the-shelf components over custom fabrication.

- Camera Assembly & Optics
 - Cameras are standard smartphone elements. While cost-effective at mass-production scales, custom manufacturing a camera module and its optical lenses for an individual prototype would cost an absurd amount of money.
 - In addition, developing a custom camera introduces unnecessary system complexity. As elaborated in section 1.4, this project prioritizes the radiofrequency, digital signal processing, hardware, and software components of the smartphone. Thus, the optical engineering required to design a custom lens system or image sensor array is explicitly excluded from the scope.
- Microchip Fabrication
 - While custom low-level designs will be implemented on programmable hardware (such as an FPGA, ASIC, or the main microcontroller), the silicon components themselves will be purchased off-the-shelf. Fabricating custom silicon or physical microchips is technically complex, requires highly specialized semiconductor facilities, and is financially prohibitive for an individual project.
 - This exclusion also applies to peripheral integrated circuits (ICs) and modules, such as the cellular baseband module, SIM card interface, and power management/charging interface. While fabrication limits still apply, the primary engineering justification here is interoperability and compliance. Using standardized, off-the-shelf components for these systems ensures seamless compatibility with industry-standard protocols (e.g., utilizing a certified USB-C receptacle guarantees physical and electrical compliance with universal chargers, minimizing the risk of system-level power delivery errors). Furthermore,

leveraging standard interfaces ensures long-term maintainability, allowing end-users to utilize ubiquitous, readily available third-party accessories for replacement and charging without needing a system-level hardware redesign if an external peripheral fails.

- Display Assembly & Touch Interface
 - This component is explicitly excluded from custom fabrication due to the large costs and manufacturing precision required to produce raw LCD or OLED panels.
- Speaker & Microphone
 - Audio transducers, including the internal speaker and microphone, will be integrated as off-the-shelf components. Fabricating custom micro-speakers or MEMS microphones requires specialized acoustic membrane manufacturing that is well outside the scope of this project. Utilizing standard components ensures known acoustic performance, predictable impedance matching for the audio amplifier circuit, and compatibility with standard audio output expectations.

1.2 - Designed components

In contrast to section 1.1, the following list defines the components that will have a custom design. Note that both 1.1 and 1.2 are not comprehensive but rather general lists.

- Chassis
 - One of the main cruxes of this project is developing a custom smartphone chassis modeled after a fruit, specifically an avocado. Rather than adhering to conventional form factors, this physical enclosure is deliberately engineered to inject a sense of whimsy and elevate the user experience through a playful appearance.
- Software & Operating System
 - The operating system will not be designed entirely from scratch. It will instead utilize a customized Linux kernel configuration, specifically modified to accommodate the non-standard geometry of the avocado-shaped display assembly.
 - Conversely, standard smartphone application software must be custom developed from the ground up. This includes core utilities such as the messaging client, web browser, and local mobile applications, all engineered to natively support the unique user interface.
- Printed Circuit Board & Electrical Routing
 - This custom smartphone architecture requires a completely unique layout, meaning a standard development board with pre-connected components is insufficient. The main system board (PCB) must be custom-designed and routed to fit the physical constraints of the avocado-shaped chassis while establishing the necessary electrical interconnections between all selected subsystems.

- Low-Level System Logic
 - The FPGA and microcontrollers will utilize custom low-level logic to implement firmware functions specific to the operational requirements of the avocado design. This local processing architecture will handle critical hardware-level tasks, including custom cryptographic encryption algorithms, thermal management routines to prevent overheating, and peripheral bus communication.

1.3 Main design tools

- Altium Designer: Schematic capture and multi-layer PCB layout routing
- SolidWorks / Onshape: 3D mechanical design and enclosure stacking for the chassis
- Xilinx Vivado: HDL simulation, synthesis, and programming in Verilog.
- Visual Studio Code: Bare-metal C development for microcontrollers and firmware
- Qt Creator: Native C++ cross-compilation and QML user interface design)

1.4 Project priorities

This project is primarily for educational purposes. However, there are specific sectors of engineering that will be focused on, as shown by the choices of off-the-shelf and custom designed aspects of the smartphone. The following list details what the main learning areas are.

- Radio Frequency (RF): Understanding RF principles in relation to telecommunications, and in this context, related to how smartphones interface with cellular networks.
- Digital Signal Processing (DSP): Understanding signal processing and its applications within smartphones.
- Hardware: Learning how to design PCBs and electrical routing that allow for the entire device to function optimally.
- Software: Learning to create user-friendly UI that adapts to non-standard display geometry to make using the device as straightforward as possible.

2. Technical objectives

This section outlines the technical objectives for this smartphone project, or in other words, what the device should be able to do.

2.1 General abilities

The resulting phone should be able to do the following basic tasks:

- Send SMS messages to other devices
- Access the internet and play a YouTube video (rendering audio, video, and text)
- Take pictures.
- Play a specialized version of Piano Tiles (developed specifically for the device).

- Maintain a basic level of security by demonstrating resilience against standard modern malware.

2.2 Precise abilities

- Photo and video rendering should be up to 720p (both on the device and from external sources)
- Wi-Fi connection signal strength should reach up to -70 dBm
- SMS messages should take no longer than 30 seconds to be sent and received
- Idle standby time of at least 24 hours
- Continuous active runtime of at least 2 hours on a single full charge
- Boot time should be no longer than 45 seconds
- Touch input response no longer than 100 ms

3. Project timeline

This section defines the project timeline, and target dates for large, high-level tasks. Smaller subtasks will be managed as time goes on; however, this schedule remains an overview so that the project remains on track.

Year	Month	Tasks to be accomplished
2026	October	- Preliminary research - High-level hardware circuit design and component selection
	November	- PCB circuit design - High level FPGA & microcontroller logic design
	December	- FPGA & microcontroller full logic design - Basic software & OS development (SMS messaging, web browser, Linux kernel)
2027	January	- PCB fabrication - Supplementary software development (games, aesthetics) - Chassis re-design and creation
	February	- Testing & fixing
	March	- Testing & fixing - Presentation of final product

4. Finalized budget

The total finalized budget for this project is \$1,000 CAD. Table 2 how this budget will be spent per category:

Category	Budget (\$ CAD)
Software (simulation tools, IDEs, etc)	0 with student licenses
PCB (without FPGA chip)	700
FPGA chip	200
Chassis	100